

James Conlon

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EDUCATION

Boston University

B.S. in Computer Engineering, Minor in Computer Science (GPA: 3.56)

Boston, MA

Sep 2022 – May 2026

EXPERIENCE

Embedded Software and Security Intern

June 2025 – Present

ei³ : Industrial Internet Intelligence

Pearl River, NY

- Designed and implemented a novel OpenVPN-based TAP server solution using Linux network namespaces, policy routing, and multiple “virtual proxy” VPN clients to resolve client-side NAT and subnet overlap issues for legacy industrial devices.
- Developed a comprehensive Python migration service to automate industrial router (Ewon, mGuard) configuration by downloading, validating, and uploading new profiles, ensuring zero-downtime migration while parsing and preserving full customer network topologies.
- Collaborated with internal support teams to author the SOP for automated remote deployments, establishing a safe migration process that preserves complex customer network topologies like VLANs and static routes.
- Designed and owned the REST API and data model (Python/Flask) for managing device configurations and firmware, allowing future expansion for new device models or new device-specific features.

Embedded Software and Security Intern

June 2024 – June 2025

ei³ : Industrial Internet Intelligence (FT summer, PT during semesters)

Pearl River, NY

- Engineered a configurable OpenVPN TUN server in Docker, designed for deployment on edge hardware to establish a migration path for third-party industrial devices onto the ei³ platform.
- Simulated factory networks (OPC-UA, MODBUS, PLCs), and performed competitive benchmarking against key competitors to evaluate technical capabilities, user experience, and security.

IT Support Specialist

Aug 2022 – Dec 2024

BU Information Services and Technology

Boston, MA

- Diagnosed and resolved hardware/software issues: Windows/Mac/Linux, networking, and academic software.

PROJECTS

Predictive Maintenance Edge Data Logger | C, Python, AWS S3, Tailscale, BeagleBone

May 2025

- Developed an end-to-end connected edge device data logger on a BeagleBone Black, using C for data acquisition from an OPC-UA server and Python for ML and cloud integration.
- Designed a dual-rate logging system to minimize cloud costs, capturing high-frequency data locally for ML model training and low-frequency data for long-term storage in AWS S3.
- Implemented secure remote access using Tailscale, air-gapping a factory network while enabling remote device management and data retrieval.
- Deployed a local Python ML model for predictive maintenance, analyzing high-frequency data to forecast equipment failures and automatically send email alerts to operators.

FPGA-Based Keyboard Synthesizer | Verilog, FPGA, Digital Signal Processing, Team Project

Dec 2023

- Developed a digital synthesizer on Artix-7 FPGA, enabling simulation of musical notes and octaves.
- Designed a module for generating PWM signals with variable duty cycles to simulate sound waveforms including square, sine, and triangle waves.
- Integrated PS/2 keyboard input for control, along with a 7-segment display and audio output for feedback.
- Collaborated with team to address challenges such as sound generation and timing issues related to waveform indexing.

TECHNICAL SKILLS

Languages: Python, Go, C (systems-level), Bash, Verilog

Framework & Tools: Flask, Linux (Debian/Alpine/RHEL/Fedora), OpenSSL

DevOps & Virtualization: Docker, Docker Compose, Git, CI/CD (GitHub Actions), Jira

Networking and Security: TCP/IP, OpenVPN, NAT, VLANs, Policy Routing, PKI, Wireshark

Courses: Systems: {Distributed, Embedded, Operating}, Cybersecurity, Client-Server Software, Intro Databases